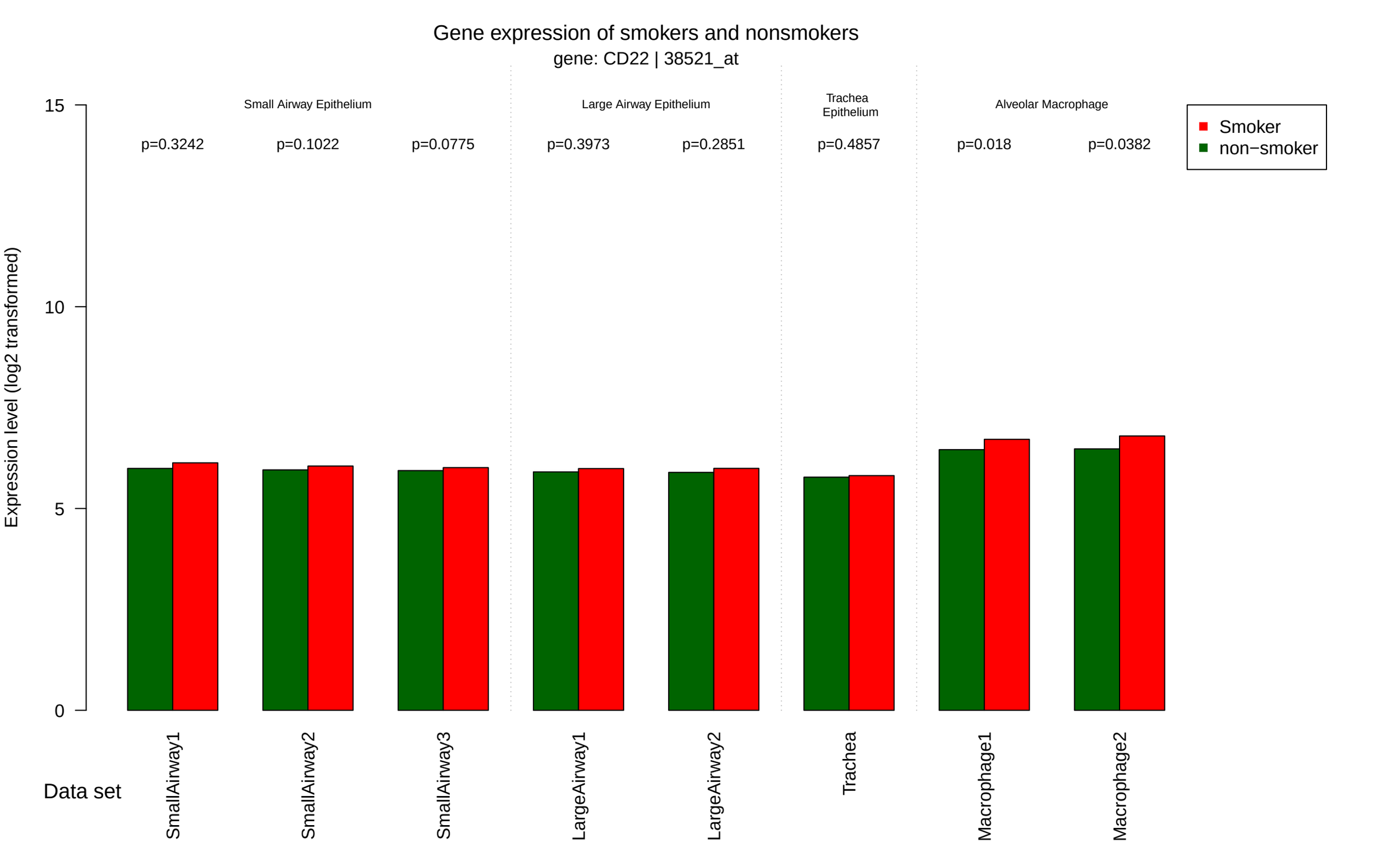
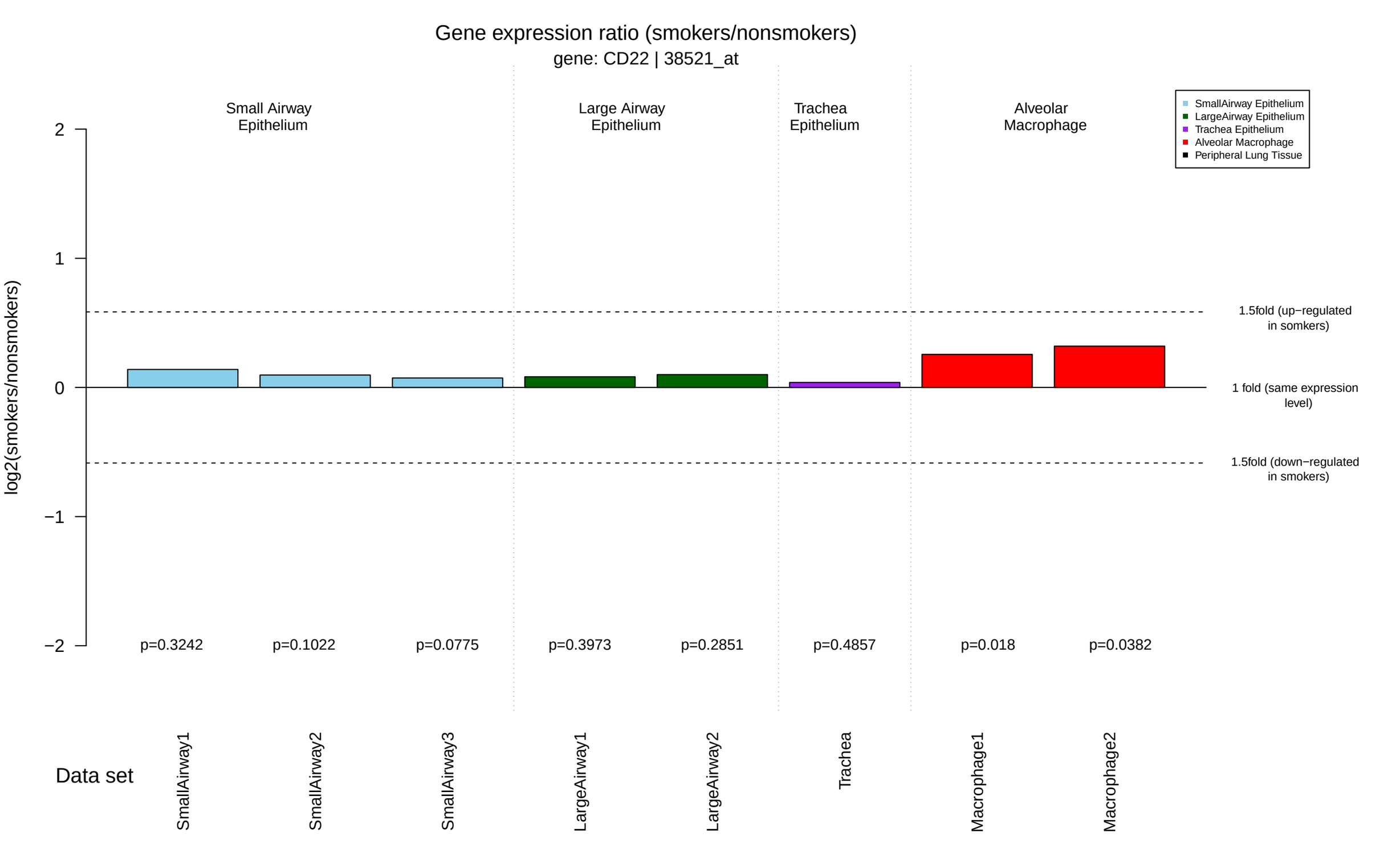
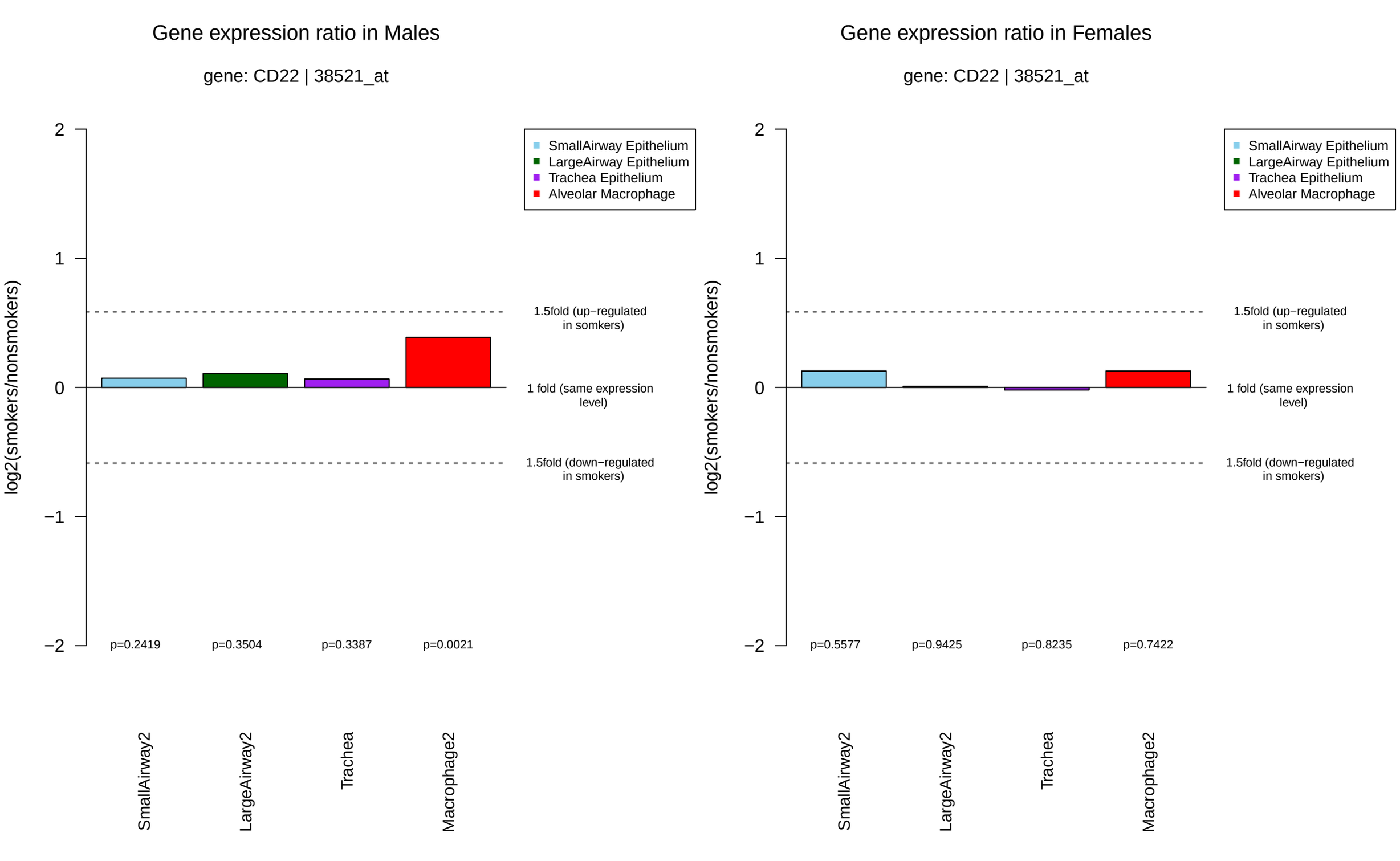
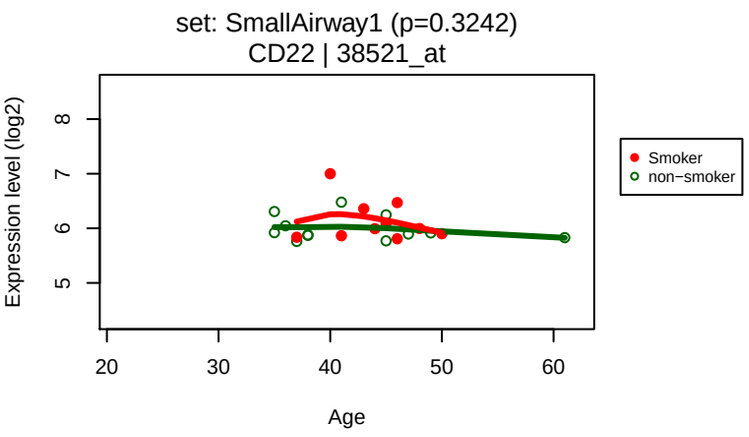
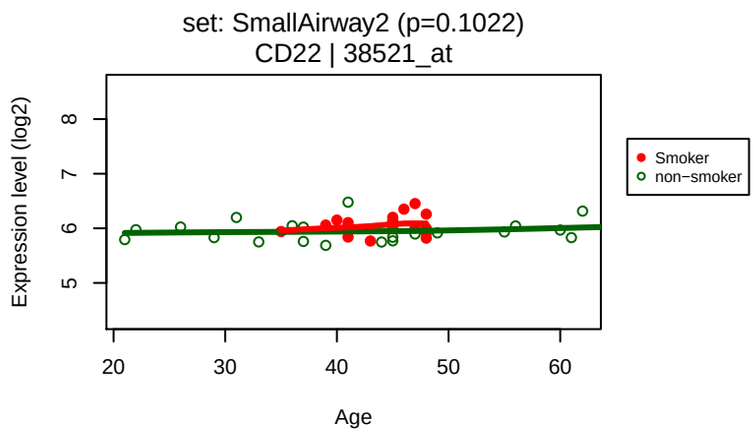
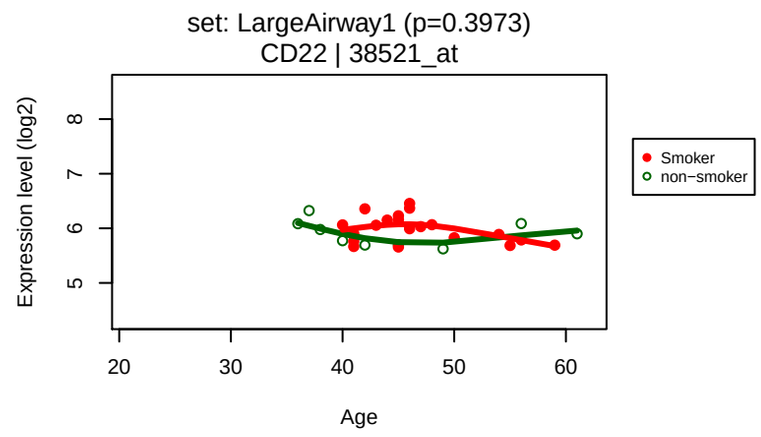

Gene expression of smokers and nonsmokers
gene: CD22 | 38521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3242
p=0.1022
p=0.0775
p=0.3973
p=0.2851
p=0.4857
p=0.018
p=0.0382
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

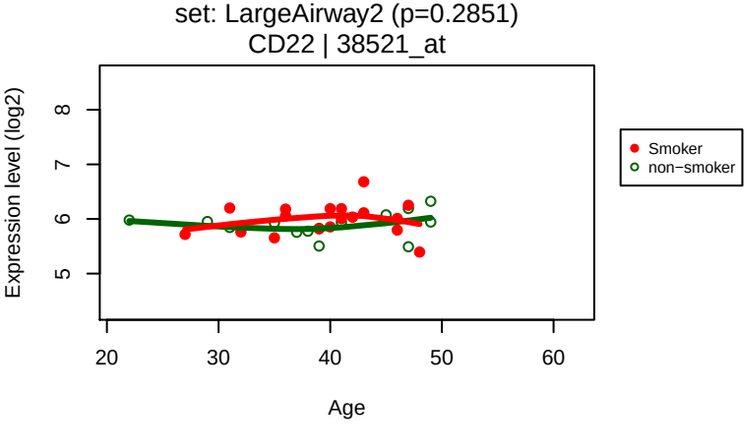

Gene expression ratio (smokers/nonsmokers)
gene: CD22 | 38521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3242
p=0.1022
p=0.0775
p=0.3973
p=0.2851
p=0.4857
p=0.018
p=0.0382
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

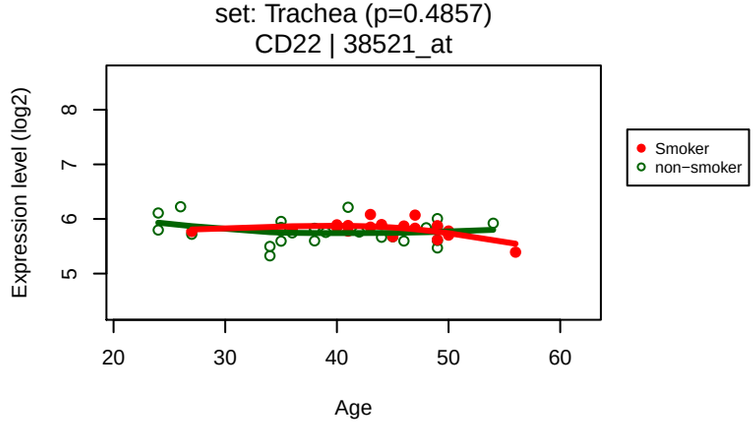

Gene expression ratio in Males
gene: CD22 | 38521_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2419
p=0.3504
p=0.3387
p=0.0021
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CD22 | 38521_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5577
p=0.9425
p=0.8235
p=0.7422
SmallAirway2
LargeAirway2
Trachea
Macrophage2

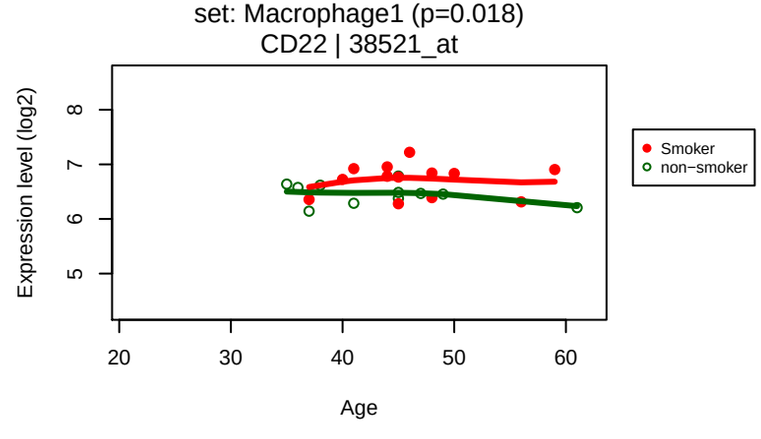

set: SmallAirway1 (p=0.3242)
CD22 | 38521_at
Expression level (log2)
Age
Smoker
non−smoker

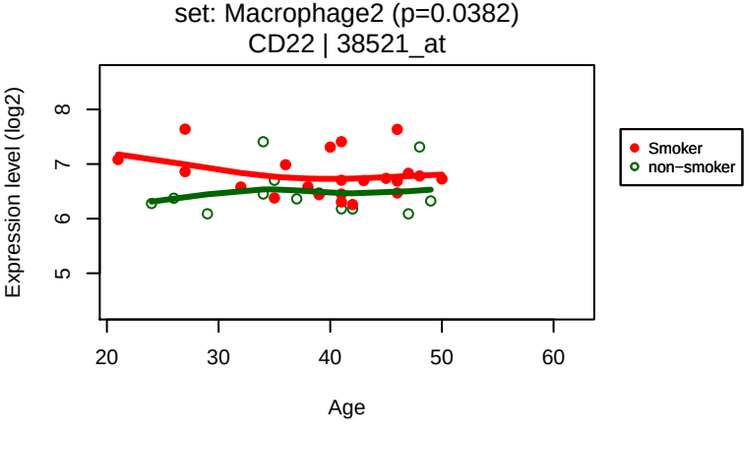

set: SmallAirway2 (p=0.1022)
CD22 | 38521_at
Expression level (log2)
Age
Smoker
non−smoker

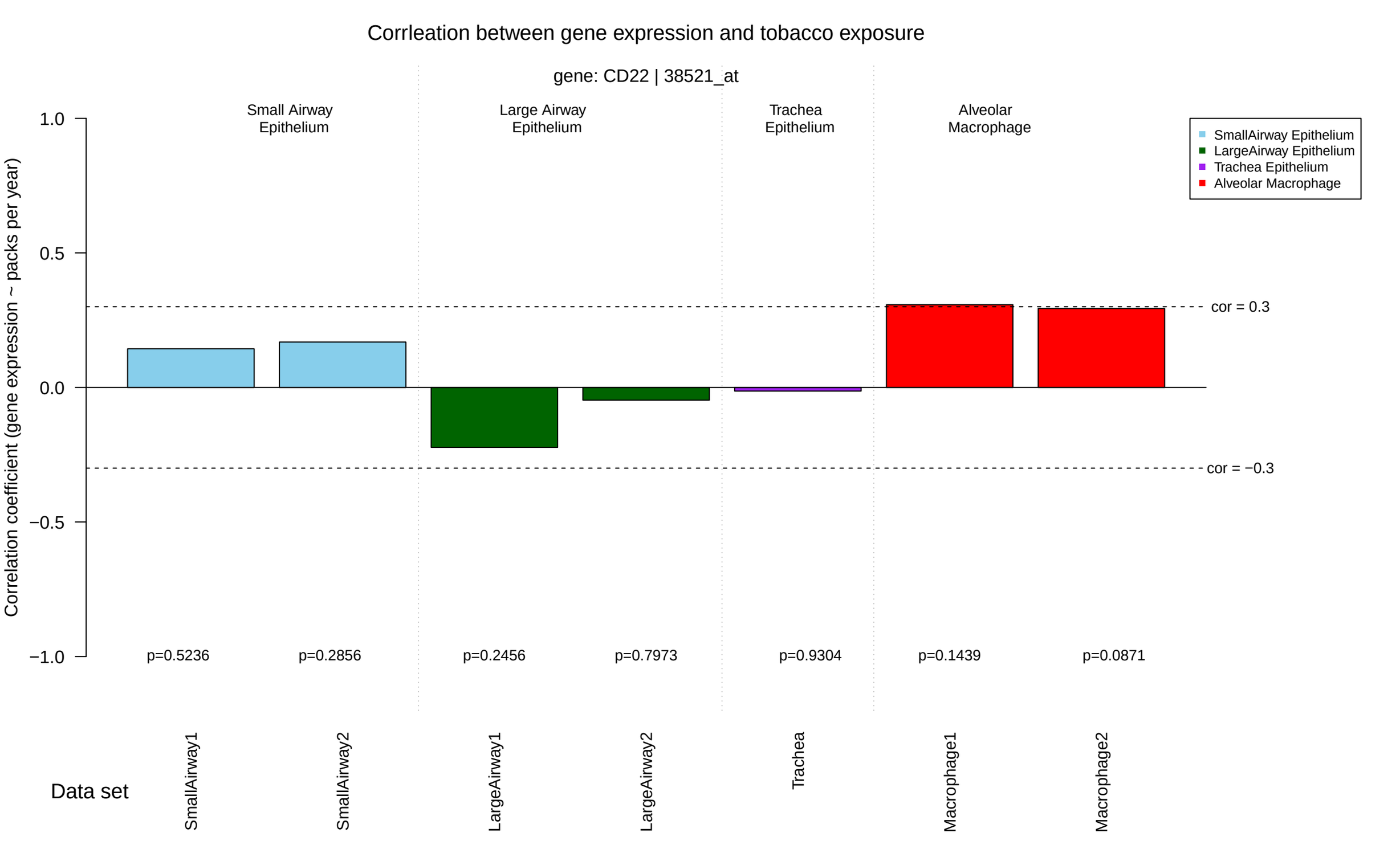

set: LargeAirway1 (p=0.3973)
CD22 | 38521_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2851)
CD22 | 38521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4857)
CD22 | 38521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.018)
CD22 | 38521_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0382)
CD22 | 38521_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CD22 | 38521_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5236
p=0.2856
p=0.2456
p=0.7973
p=0.9304
p=0.1439
p=0.0871
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2